

4.Git-ConflictResolution

GitLab Link:

<https://gitlab.com/codebydeshma-27-group/CodebyDeshma-27-project>

Experiment Name: Git Conflict Resolution

Part A – Creating Branch and File Changes

Step 1: Verify Master Branch Status

Command

git status **Output**

```
deshm@Deshma MINGW64 ~/GitDemo (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.
nothing to commit, working tree clean
```

Step 2: Create New Branch

Command

git branch GitWork **Output**

```
deshm@Deshma MINGW64 ~/GitDemo (main)
$ git branch GitWork
```

Step 3: Switch to GitWork Branch

Command

git checkout GitWork

Output

```
deshm@Deshma MINGW64 ~/GitDemo (main)
$ git checkout GitWork
Switched to branch 'GitWork'
```

Step 4: Create hello.xml File in Branch

Command

echo "<h1>Hello from Branch</h1>" > hello.xml

Output

```
deshm@Deshma MINGW64 ~/GitDemo (GitWork)
$ echo "<h1>Hello from Branch</h1>" > hello.xml
```

Step 5: Check Git Status

Command

git status **Output**

```
deshm@Deshma MINGW64 ~/GitDemo (GitWork)
$ git status
On branch GitWork
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hello.xml

nothing added to commit but untracked files present (use "git add" to track)
deshm@Deshma MINGW64 ~/GitDemo (GitWork)
$
```

```
deshm@Deshma MINGW64 ~/GitDemo (GitWork)
$ cat hello.xml
<h1>Hello from Branch</h1>
```

Step 6: Add and Commit Changes in Branch

Commands

git add hello.xml

git commit -m "Added hello.xml in branch" **Output**

```
deshm@Deshma MINGW64 ~/GitDemo (GitWork)
$ git add hello.xml
warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next time Git touches it

deshm@Deshma MINGW64 ~/GitDemo (GitWork)
$ git commit -m "Added hello.xml in main"
[GitWork 1add749] Added hello.xml in main
1 file changed, 1 insertion(+)
create mode 100644 hello.xml
```

Part B – Checking Differences

Step 7: View Git Logs

Command

git log --oneline --graph --decorate --all **Output**

```
deshm@Deshma MINGW64 ~/GitDemo (GitWork)
$ git log --oneline --graph --decorate --all
* 1add749 (HEAD -> GitWork) Added hello.xml in main
* 7c1dab2 (origin/main, main) Added branch file
* daa7ce7 Updated gitignore rules
* f151e69 Added gitignore file
* ccfbf47 Merge branch 'main' of https://gitlab.com/codebydeshma-27-group/CodebyDeshma-27-project
| \
|  * 474fda1 Initial commit
|  * 69c13cc Added welcome.txt file
```

Step 8: Compare Differences Using Git Diff

Command

git diff master GitWork **Output**

```
deshm@Deshma MINGW64 ~/GitDemo (GitWork)
$ git diff main GitWork
diff --git a/hello.xml b/hello.xml
new file mode 100644
index 0000000..915034f
--- /dev/null
+++ b/hello.xml
@@ -0,0 +1 @@
+<h1>Hello from Branch</h1>
```

Part C – Merge Conflict Resolution

Step 9: Merge Branch with Master

Command

git merge GitWork **Output**

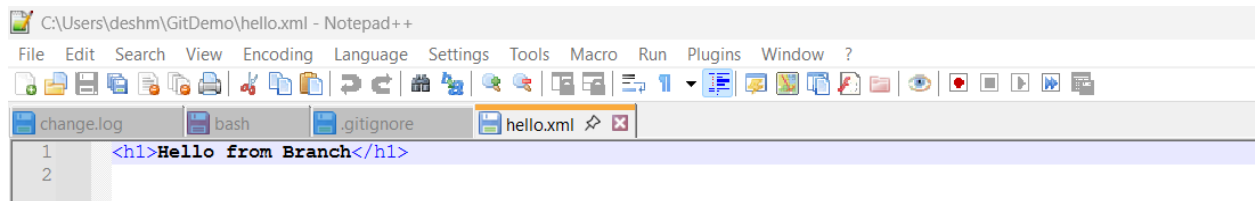
```
deshm@Deshma MINGW64 ~/GitDemo (GitWork)
$ git merge GitWork
Already up to date.
```

Step 10: Observe Conflict Markup in hello.xml

Command

notepad++ hello.xml **Output**

```
deshm@Deshma MINGW64 ~/GitDemo (GitWork)
$ notepad++ hello.xml
```



Step 11: Commit Resolved Changes

Commands

git add hello.xml

git commit -m "Resolved merge conflict" **Output**

```
deshm@Deshma MINGW64 ~/GitDemo (GitWork)
$ git add hello.xml

deshm@Deshma MINGW64 ~/GitDemo (GitWork)
$ git commit -m "Resolved merge conflict"
On branch GitWork
nothing to commit, working tree clean
```

Part D – Updating .gitignore

Step 12: Add Backup File to .gitignore

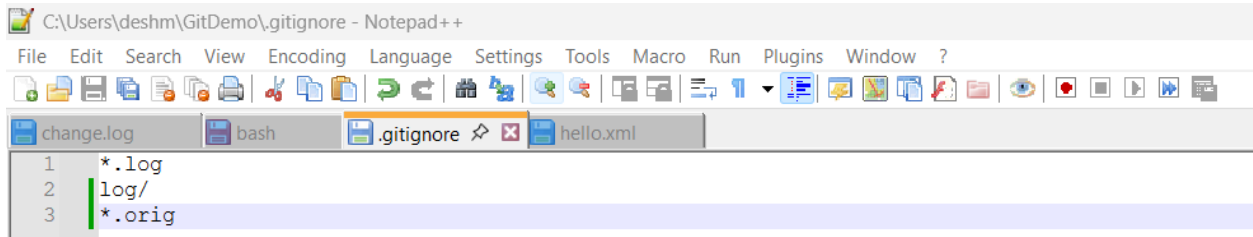
Command

notepad++ .gitignore

Content Added

*.orig **Output**

```
deshm@Deshma MINGW64 ~/GitDemo (GitWork)
$ notepad++ .gitignore
```



Step 13: Commit .gitignore Changes

Commands

git add .gitignore

git commit -m "Updated gitignore" **Output**

```
deshm@Deshma MINGW64 ~/GitDemo (GitWork)
$ git add .gitignore

deshm@Deshma MINGW64 ~/GitDemo (GitWork)
$ git commit -m "Updated gitignore"
[GitWork c1de9a5] Updated gitignore
1 file changed, 2 insertions(+), 1 deletion(-)
```

Part E – Branch Cleanup and Verification

Step 14: List All Branches Command

git branch **Output**

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```
deshm@Deshma MINGW64 ~/GitDemo (GitWork)
$ git branch
* GitWork
main
```

Step 15: Delete Merged Branch

Command

git branch -d GitWork **Output**

```
deshm@Deshma MINGW64 ~/GitDemo (GitWork)
$ git branch -d GitWork
error: cannot delete branch 'GitWork' used by worktree at 'C:/Users/deshm/GitDemo'
```

Step 16: Verify Final Git Logs

Command

git log --oneline --graph --decorate **Output**

```
deshm@Deshma MINGW64 ~/GitDemo (GitWork)
$ git log --oneline --graph --decorate
* c1de9a5 (HEAD -> GitWork) Updated gitignore
* 1add749 Added hello.xml in main
* 7c1dab2 (origin/main, main) Added branch file
* daa7ce7 Updated gitignore rules
* f151e69 Added gitignore file
* ccfbf47 Merge branch 'main' of https://gitlab.com/codebydeshma-27-group/CodebyDeshma-27-project
|
| * 474fda1 Initial commit
| * 69c13cc Added welcome.txt file
```

Result

Thus, the merge conflict was resolved successfully using Git merge tools and the changes were committed successfully.